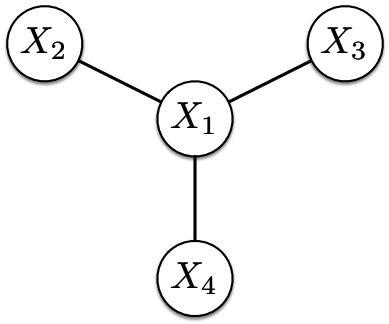
Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 9 Solutions

Problem 9.1

Let $X_1, X_2, X_3, X_4 \in \mathcal{X} = \{0, 1\}$ be binary random variables and consider the following tree graphical model representing the structure in the joint distribution p_{X_1,X_2,X_3,X_4}:



Moreover, suppose

$$\phi_{1,2}(x, x') = \phi_{1,3}(x, x') = \phi_{1,4}(x, x') = \begin{cases} 3 & x = 0,\, x' = 1 \\ 1 & \text{otherwise,} \end{cases}$$
$$\phi_2(x) = \phi_3(x) = \phi_4(x) = 1, \qquad x \in \mathcal{X},$$
$$\phi_1(x) = \begin{cases} \alpha & x = 1 \\ 1 & x = 0 \end{cases}$$

where $\alpha > 0$ is a constant.

(a) Compute all sum-product algorithm messages.

Solution: We are free to choose any node as the root node, so we arbitrarily pick the node associated with X_1. Then the leaf-to-root messages are

$$m_{2\to1}(x_1) = \sum_{x_2} \phi_2(x_2)\, \phi_{1,2}(x_1, x_2) = \begin{cases} 4 & x_1 = 0 \\ 2 & x_1 = 1, \end{cases}$$

and, by symmetry (i.e., all edge potentials are the same and all node potentials for leaf nodes are the same),

$$m_{3\to1}(x_1) = m_{4\to1}(x_1) = m_{2\to1}(x_1), \qquad x_1 \in \mathcal{X}.$$

In turn, the root-to-leaf messages are

$$m_{1\to 2}(x_2) = \sum_{x_1} \phi_1(x_1)\, \phi_{1,2}(x_1, x_2)\, m_{3\to 1}(x_1)\, m_{4\to 1}(x_1)$$
$$= \begin{cases} 1 \cdot 1 \cdot 4 \cdot 4 + \alpha \cdot 1 \cdot 2 \cdot 2 = 16 + 4\alpha & x_2 = 0 \\ 1 \cdot 3 \cdot 4 \cdot 4 + \alpha \cdot 1 \cdot 2 \cdot 2 = 48 + 4\alpha & x_2 = 1, \end{cases}$$

and, again by symmetry,

$$m_{1\to 3}(x) = m_{1\to 4}(x) = m_{1\to 2}(x), \qquad x \in \mathcal{X}.$$

(b) Compute all marginal distributions.

Solution: We obtain the root marginal by multiplying the messages as follows:

$$p_{X_1}(x_1) \propto \phi_1(x_1)\, m_{2\to 1}(x_1)\, m_{3\to 1}(x_1)\, m_{4\to 1}(x_1) = \begin{cases} 64 & x_1 = 0 \\ 8\alpha & x_1 = 1, \end{cases}$$

i.e.,

$$p_{X_1}(x_1) = \begin{cases} 8/(8+\alpha) & x_1 = 0 \\ \alpha/(8+\alpha) & x_1 = 1. \end{cases}$$

For the the leaf marginals we have

$$p_{X_2}(x_2) \propto m_{1\to 2}(x_2),$$

i.e.,

$$p_{X_2}(x_2) = \begin{cases} (4+\alpha)/(16+2\alpha) & x_2 = 0 \\ (12+\alpha)/(16+2\alpha) & x_2 = 1, \end{cases}$$

and, once again by symmetry,

$$p_{X_3}(x) = p_{X_4}(x) = p_{X_2}(x), \qquad x \in \mathcal{X}.$$

Note that since the alphabet is of cardinality 2 for all variables, our edge potentials are represented by tables of size 2×2, our messages are tables of size 2×1, and our marginals are represented by 1×1 tables, i.e., scalars.

(c) Determine the range of values α for which $p_{X_i}(1) > p_{X_i}(0)$, i.e., X_i is more likely to be 1 than 0, $i = 1, 2, 3, 4$.

Solution: To obtain the answer for the root, we use the expressions for the marginal distributions in part (b) to solve the following inequality:

$$p_{X_1}(1) > p_{X_1}(0)$$
$$\alpha > 8,$$

i.e., for $\alpha > 8$, X_1 is more likely to be equal to 1. Note the trade-off between the node potential that for $\alpha > 1$ favors configurations in which $X_1 = 1$ and the edge potential that favors configurations in which $X_1 = 0$. Evidently, for $\alpha > 8$, the influence of the node potential outweighs that of the edge potential.

We proceed similarly for the leaf nodes and obtain for $i = 2, 3, 4$

$$\begin{aligned} p_{X_i}(1) &> p_{X_i}(0) \\ 12 + \alpha &> 4 + \alpha \\ 8 &> 0, \end{aligned}$$

i.e., for any $\alpha > 0$, the leaf node is more likely to be equal to 1. This is not surprising as the edge potential favors configurations with the leaf nodes equal to one and the node potential is uniform.